

Week 9

Control

Jaerock Kwon, PhD

Associate Professor

Electrical and Computer Engineering

University of Michigan-Dearborn

Two loops, opposite ends of the vehicle

	Inner	Outer
Job	Hold a commanded steering angle	Follow a path
Runs on	Teensy 4.1	Laptop
Rate	200 Hz	~20 Hz
Method	PID	Pure pursuit

They have almost nothing in common except the word **control**.

1. PID

What each term buys, and costs

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de}{dt}$$

Term	Buys	Costs
P	Speed of response	Steady-state offset; oscillation
I	Kills steady-state offset	Windup, overshoot, sluggishness
D	Damping	Amplifies sensor noise

D on a noisy encoder cooks a gearmotor

Differentiating a noisy signal amplifies the noise.

On a mechanical steering column that is not just **seen** — it is **actuated**.

Audible chatter, and real current draw.

- Filter the derivative
- Or start at PD with a small D
- Or use derivative-on-**measurement**, so a setpoint step does not produce a spike

Where the loop lives

Not in ROS. Two reasons.

Timing. Linux cannot promise 200 Hz. A loop that **usually** makes its deadline oscillates unpredictably under load.

Survival. If the USB link drops, a loop running in ROS is **gone**.

A loop on the Teensy is still there — and can centre the wheels and de-energise, which is what failsafe row 1 requires.



If the loop cannot hold $\leq 1^\circ$ steady-state with no sustained oscillation at Stage 1, adopt the daughterboard **rather than continuing to tune.**

E4 — written **before** anyone started tuning.

Why pre-registration matters

The decision to stop tuning is otherwise made by someone who is **three days into tuning** and emotionally invested in it.

Electrical track: $\leq 1^\circ$ is your number, and Merge 1 is where it fires.

Invoking E4 costs you nothing here — the record already calls it the planned action.

2. Path tracking

A different problem entirely

There is no setpoint in the same sense.

There is a **path**, and the vehicle is somewhere near it.

Pure pursuit

Pick a point on the path a lookahead distance L_d ahead.

Compute the arc from here to there.

$$\kappa = \frac{2 \sin \alpha}{L_d} \quad \delta = \arctan(\kappa L)$$

- No error term
- No gains
- No integrator

Pure geometry — which is why it cannot ask for a manoeuvre the car cannot perform.

Lookahead is the one real knob

L_d	Behaviour
Too short	Tracks tightly, oscillates, chases every wiggle
Too long	Smooth and stable, but cuts corners

What "Regulated" adds

- **Curvature regulation** — slow down in tight turns
- **Proximity regulation** — slow down near obstacles
- **Velocity-scaled lookahead** — longer when fast, shorter when slow

```
desired_linear_vel: 0.6          # well under the 1.4 m/s cap  
lookahead_dist: 0.8  
min_lookahead_dist: 0.4  
max_lookahead_dist: 1.2  
use_velocity_scaled_lookahead_dist: true  
min_turning_radius: 1.6
```

Two parameters that cannot both be true

```
use_rotate_to_heading: false  
allow_reversing: true
```

Set both `true` and **Nav2 refuses to start**.

Rotate-to-heading means "**stop and spin to face the right way**" — which this vehicle cannot do at all.

So `false` is not a preference. It is the only legal value for a car — and it is what makes `allow_reversing` available.

3. The constraint reaches the goal checker

These look sloppy. They are geometry.

```
goal_checker:  
  xy_goal_tolerance: 0.6  
  yaw_goal_tolerance: 0.6
```

`min_turning_radius: 1.6` = $R_{\min}$ 1.52 m + margin.

A car with that radius and no in-place rotation **cannot reduce a 0.75 m pose error**. It can only loop past and retry.

0.6 m is ~40% of the turning radius — achievable in one pass.

F-N3: the car that orbited forever

At 0.25 m tolerance the vehicle arrived within 0.75 m, then circled:

```
(-4.19, -1.46) (-5.41, 0.43) (-2.28, 0.81) (-4.51, -1.35) ...
```

It never converged. It never failed.

No error. No timeout. No diagnostic.

Just a car driving in circles around a point it could not reach.



The system did not report a problem, because from its own point of view there was not one.

Weeks 1, 3, 4 — and now 9. Every component did exactly what it was configured to do.

Track work

Track	
Electrical	Close the loop on the bench. $\leq 1^\circ$, no oscillation
Chassis	Measured <code>mitt_dimensions.yaml</code> committed — two tracks are waiting
Simulation	<code>mitt_bench</code> skeleton: measure something, print a number
Software	<code>mitt_hardware</code> <code>read()/write()</code> against a recorded bag

Merge 1 is 11/23 — two weeks.

If either half will not be ready, say so at **today's** standup.